Math 152 Lecture Notes: Week 4

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0.1 Review of Area

Last week we saw that the area of a plane region could be expressed as an integral in (at least) two ways: if the region extends from $x = a$ to $x = b$, and from $y = c$ to $y = d$, then

$$\int_a^b h(x)\mathrm{d}x = A = \int_c^d w(y)\mathrm{d}y$$

where $h(x)$ is the cross sectional height at x, and $w(y)$ is the cross sectional width. [Draw picture]

We also saw how to decompose a region into simpler regions in order to find its area.

1 Taking Advantage of Symmetry

This week we start with another trick for computing areas, and some related tricks for computing integrals more generally.

1.1 Symmetry considerations for area

When a region has a line of symmetry, we can decompose it into two pieces that have equal area. We then only need to compute the area of one of the pieces, and multiply this area by 2.

Example 1. Consider the region above $y = -1$ and below $y = 1 + x^5$ and $y = 1 - x^5$. [Draw picture, mark intersection points]. The y-axis is a line of symmetry for this region, so it suffices to find the area of the region to the right of the y-axis. This is

$$\int_0^{2^{1/5}} [1 - x^5 - (-1)]\mathrm{d}x = \int_0^{2^{1/5}} [2 - x^5]\mathrm{d}x = \left[2x - \frac{x^6}{6}\right]\Bigg|_0^{2^{1/5}} = 2 \cdot 2^{1/5} - \frac{2^{6/5}}{6}$$

The total area is twice this, so is

$$4 \cdot 2^{1/5} - \frac{2^{6/5}}{3} = 2^{11/5} - \frac{2^{6/5}}{3}$$

1.2 Integrals of even and odd functions

Symmetry considerations can be applied to integrals more generally, even if we are not thinking of them as representing areas. In this case, the relevant symmetries are those of the function being integrated. In particular, if a function is even or odd, we can use this symmetry to simplify integrals.

Definition. Let f be a function.

1. f is said to be even if $f(-x) = f(x)$ for all x.

2. f is said to be odd if $f(-x) = -f(x)$ for all x.

Examples of even functions include even powers of x and the cosine function, while examples of odd functions include odd powers of x and the sine function.

Example 2. The function $f(x) = x^3 \sin(x) - x^2$ is even, because

$$f(-x) = (-x)^3 \sin(-x) - (-x)^2 = (-x^3)(-\sin(x)) - x^2 = x^3 \sin(x) - x^2 = f(x)$$

In examples like the above, we can use the following general principles to see that a function is even or odd.

1. A sum of even functions is even
2. A sum of odd functions is odd
3. A product of two even or two odd functions is even
4. A product of an even and an odd function is odd

Knowing that a function is even or odd allows us to simplify integrals over symmetric intervals:

Proposition. *Consider a function f on a symmetric interval $[-a, a]$ around 0.*

1. *If f is odd, then*
$$\int_{-a}^{a} f(x)\mathrm{d}x = 0$$
2. *If f is even, then*
$$\int_{-a}^{a} f(x)\mathrm{d}x = 2\int_{0}^{a} f(x)\mathrm{d}x$$

This can be explained in two ways. One is geometric. [Draw pictures] For an odd function, the areas to the left and right of the y-axis cancel out, leaving zero. For even functions, the total area is the sum of two equal areas. The other way to explain this uses substitution. The point is that

$$\int_{-a}^{a} f(x)\mathrm{d}x = \int_{-a}^{0} f(x)\mathrm{d}x + \int_{0}^{a} f(x)\mathrm{d}x$$

Using the substitution $u = -x$, this is

$$\int_{a}^{0} -f(-u)\mathrm{d}u + \int_{0}^{a} f(x)\mathrm{d}x = \int_{0}^{a} f(-u)\mathrm{d}u + \int_{0}^{a} f(x)\mathrm{d}x = \int_{0}^{a} f(-x)\mathrm{d}x + \int_{0}^{a} f(x)\mathrm{d}x$$

For odd functions, these terms cancel out, while for even functions, they are equal.

Example 3. Consider the function $f(x) = \sin(2x^2)\sin(x^3)$. This satisfies

$$f(-x) = \sin(2(-x)^2)\sin((-x)^3) = \sin(2x^2)\sin(-x^3) = -\sin(2x^2)\sin(x^3) = -f(x)$$

so is odd. Therefore,

$$\int_{-1}^{1} f(x)\mathrm{d}x = 0$$

1.3 Integrals of periodic functions

Another kind of symmetry of a function that we can use is periodicity.

Definition. A function f is *periodic* with period p if $f(x+p) = f(x)$ for all x. We also say that p is a period of f.

The main examples of periodic functions are the sine and cosine functions, both of which have period 2π. [Draw picture]. Functions built out of these will also be periodic.

Example 4. $f(x) = \sin(x)^2 - \frac{1}{\cos(x)+2}$ is also periodic with period 2π.

Notice that if p is a period of f, then so is np for any integer n:

$$\cdots = f(x-p) = f(x) = f(x+p) = f(x+p+p) = \dots$$

Thus functions do not have unique periods. If we need to pick one out, we usually refer to the smallest positive period, if one exists.

Example 5. For $f(x) = c$ a constant function, any number is a period of f.

Composing a function with a linear function affects its periods in a particular way.

Proposition. *If f is periodic with period p, then p/a is a period of $g(x) = f(ax+b)$ as long as a is nonzero.*

Example 6. The function $\sin(4\pi x + 1)$ has period $1/2$.

If we know that a function is periodic, we can use this to change the bounds of an integral of that function.

Proposition. *If f is periodic with period p, then for any a, b,*

$$\int_a^b f(x)\mathrm{d}x = \int_{a+np}^{b+np} f(x)\mathrm{d}x$$

for any integer n.

This can be understood as a simple application of the change of variable formula (u-substitution) using $u = x + np$. We can also understand it in terms of area. [Draw picture of sine function.]

Example 7. What is

$$\int_5^7 \sin(2\pi x)\mathrm{d}x?$$

First of all, this function has period 1, so we can subtract 6 from both bounds of integration, leaving

$$\int_{-1}^{1} \sin(2\pi x)$$

Then, we can use the fact that the function is odd to conclude that the integral is 0.

Example 8. What is

$$\int_{3\pi/2}^{5\pi/2} \sin(x)^2\cos(x)\mathrm{d}x?$$

There are lots of ways to evaluate this; one thing we can do is shift the bounds of integration by 2π, yielding

$$\int_{-\pi/2}^{\pi/2} \sin(x)^2\cos(x)\mathrm{d}x$$

then using the fact that the function is even to replace this with

$$2\int_0^{\pi/2} \sin(x)^2\cos(x)\mathrm{d}x$$

and then use the substitution $u = \sin(x)$ to replace this with

$$2\int_0^1 u^2\mathrm{d}u = 2/3$$

2 Volume

We will now return to geometric applications of the integral, moving from areas of plane regions to volumes of solids. For us, a solid is a region in 3 dimensional space bounded by surfaces.

The basic principle in our study of volume is a formula for the volume of a simple kind of solid. Given a plane region Ω, an *extrusion*[1] of Ω with height h is the solid swept out by Ω as it moves a distance h perpendicular to the plane. [Draw picture]

Proposition. *If Ω has area A, then an extrusion of Ω with height h has volume $A \cdot h$.*

Example 9. A cylinder with radius r and height h has volume $\pi r^2 h$.

Remark. You might reasonably ask at this point how we can understand the classic formula πr^2 for the area of a circle in terms of our methods of computing areas of plane regions: this comes out to

$$\int_{-r}^{r} 2\sqrt{r^2 - x^2}\mathrm{d}x = \pi r^2$$

Using the substitution $u = x/r$, this reduces to the $r = 1$ case:

$$\int_{-1}^{1} 2\sqrt{1 - u^2}\mathrm{d}u = \pi$$

for the unit circle. We do not have an antiderivative for $\sqrt{1-u^2}$, so we cannot yet explain this integral, but later on we will be able to explain it using trigonometric substitution.

Example 10. A rectangular prism with dimensions x, y, z is an extrusion of a rectangle of area xy with height z, so has volume xyz.

2.1 Volume as the Integral of Cross Sectional Area

An extrusion is a solid such that along some axis, all of its cross sections are the same. For more general solids, we will consider how its cross sections vary along an axis, and approximate its volume using a sum of volumes of extrusions of these cross sections.

Example 11. In explaining the general formula for volume, it will be useful to consider a concrete example. Our example will be a cone of height 1 and base radius 1. [Draw picture]. If we slice this with planes $y = a$ for $0 \leq a \leq 1$, the cross section is a circle of radius $1 - a$. Given a partition of the interval $[0, 1]$ into

[1] The book uses the term "right cylinder with cross section Ω" for this. I use the word "extrusion" because that is what SketchUp or a CAD program would call it.

sub-intervals $[y_{i-1}, y_i]$, we can approximate the volume of the cone by the sum of volumes of cylinders of radius $r_i = 1 - y_i$ and height $y_i - y_{i-1} = \Delta y_i$; [draw step picture]

$$\sum_i \pi(1-y_i)^2 \cdot \Delta z_i$$

The cylinders are extrusions of the cross section at y_i. Taking finer and finer partitions, these approximations get closer and closer to the true volume on one hand, and to the integral

$$\int_0^1 \pi(1-y)^2 \mathrm{d}y$$

on the other. Therefore, the integral and the true volume coincide. Evaluating this using $u = 1 - y$, we get

$$-\int_1^0 \pi u^2 \mathrm{d}u = \int_0^1 \pi u^2 \mathrm{d}u = \left[\frac{\pi u^3}{3}\right]\Big|_0^1 = \frac{\pi}{3}$$

for the volume of the cone.

The same argument can be applied to more general solids. [Draw picture] Suppose that we have chosen an axis, let x represent a position along the axis. We can slice our solid with a plane perpendicular to the axis at position x, yielding a cross section which is a plane region with some area $A(x)$. For a partition of the interval over which the solid lives, a Riemann sum

$$\sum_i A(x_i)\Delta x_i$$

represents the volume of a collection of extrusions approximating the solid. Taking finer and finer partitions to approach the true value, we have the following conclusion:

Proposition. *If $A(x)$ represents the cross-sectional area of a solid living between $x = a$ and $x = b$, then the volume is given by the integral*

$$V = \int_a^b A(x)\mathrm{d}x.$$

With this integral expression, we can rederive some formulas for volumes of familiar solids.

Example 12. Let's compute the volume of a sphere of radius r. [Draw picture with axes] We can take cross sections of the sphere in planes of fixed x-value, so x will be our variable of integration. The cross section at x is a circle of radius $\sqrt{r^2 - x^2}$ [draw picture with right triangle], and so has are $\pi(r^2 - x^2)$. Therefore, the volume is

$$\int_{-r}^r \pi(r^2 - x^2)\mathrm{d}x$$

This is an even function, so this is

$$2\int_0^r \pi(r^2 - x^2)\mathrm{d}x = 2\pi\left[r^2 x - \frac{x^3}{3}\right]\Big|_0^r = \frac{4\pi r^3}{3}$$

Example 13. Let's compute the volume of a square pyramid with base side length ℓ and height h. [Draw picture] We use the y-axis. The cross section at height y is a square of side length $\ell(1 - y/h)$ [Picture]. Thus the volume is

$$\int_0^h \ell^2(1-y/h)^2 \mathrm{d}y$$

Taking $u = 1 - y/h$, this comes out to

$$-h\ell^2 \int_1^0 u^2 \mathrm{d}u = h\ell^2 \int_0^1 u^2 \mathrm{d}u = \frac{h\ell^2}{3}$$

Example 14. For a less familiar example, consider a solid whose base is a circle of radius 1 and whose cross sections perpendicular to the x-axis are equilateral triangles [draw picture.] First of all, using the Pythagorean Theorem, one can show that the area of an equilateral triangle with side length ℓ is $\sqrt{3}\ell^2/4$. The side length of the cross section at x is $\ell(x) = \sqrt{1-x^2}$ [draw picture] so the volume comes out to

$$\int_{-1}^1 \frac{\sqrt{3}}{4}(1-x^2)\mathrm{d}x = 2\int_0^1 \frac{\sqrt{3}}{4}(1-x^2)\mathrm{d}x = \frac{\sqrt{3}}{2}\int_0^1 [1-x^2]\mathrm{d}x = \frac{\sqrt{3}}{2}\left[x - \frac{x^3}{3}\right]\Big|_0^1 = \frac{\sqrt{3}}{2}\cdot\frac{2}{3} = \frac{1}{\sqrt{3}}$$

2.2 Solids of Revolution

In order to apply find the volume of a solid by the method just introduced, we need a formula for the cross sectional area when slicing perpendicular to some axis. One case in which one can find a nice formula is if the solid is obtained by rotating a plane region about an axis.

Definition. Given a plane region Ω and an axis ℓ (that is, a line) in the plane, we obtain a *solid of revolution* by rotating the plane about ℓ and letting Ω sweep out a solid. [Draw two pictures, one where the regions abuts the axis and another where there is a separation]

Any solid with full rotational symmetry, like a cylinder, sphere, or cone, can be obtained in this way. [Draw picture] In such cases, the natural axis to use to compute the volume is the axis of rotational symmetry; the cross sections perpendicular to this axis will be particularly simple shapes. For simplicity let's take the axis of rotation to be the x-axis for now. If the region we are revolving is the region between the x-axis and the graph of a function $f(x)$, then the cross section at x is a disk of radius $f(x)$. [Draw picture.] Therefore, the volume is given by the following formula:

Proposition. *For the solid obtained by revolving the area between the x-axis and the graph of f over the interval $[a, b]$ around the x-axis, its volume is*

$$V = \int_a^b \pi f(x)^2 \mathrm{d}x$$

The same is true for solids of revolution around any axis. [Draw picture, with formula.]

Example 15. Consider the horn shape obtained by revolving the area below the graph of $\frac{1}{x}$ over $[1, 3]$ about the x axis. [Draw picture.] Applying the formula, this has volume

$$\int_1^3 \frac{\pi}{x^2}\mathrm{d}x = \left[-\frac{\pi}{x}\right]\Big|_1^3 = \frac{2\pi}{3}$$

If the plane region we are revolving does not meet the axis of rotation, then the cross sections will not be disks. We will consider the particular case of revolving a region between the graphs of two functions $0 < f(x) < g(x)$ over an interval $[a, b]$ about the x-axis. [Draw picture.] In this case, the cross section will be a *washer*[2] [draw picture] with inner radius $f(x)$ and outer radius $g(x)$. The area of such a shape is the difference between the areas of two circles: $\pi(g(x)^2 - f(x)^2)$.

[2] also called annulus, plural annuli

Proposition. *For the solid obtained by rotating the region bounded between the graphs of f and g over $[a, b]$ about the x-axis, its volume is*

$$\int_a^b \pi(g(x)^2 - f(x)^2)\mathrm{d}x$$

As before, any axis can play the role of the x-axis here.

Remark. Using the linearity of the integral, we can also understand this formula as a difference of two volumes: the volume enclosed by the outer surface minus a "core" enclosed by the inner surface.

Example 16. Consider the solid obtained by revolving the region between the graphs of $x = 1 + y$ and $x = 3$ and between $y = 0$ and $y = 2$ about the y-axis. [Draw picture]. The cross sections are washers with inner radius $1 + y$ and outer radius 3, so the volume is

$$\int_0^2 \pi(3^2 - (1+y)^2)\mathrm{d}y = 18\pi - \pi\int_0^2 (1+y)^2\mathrm{d}y = 18\pi - \pi\int_1^3 u^2\mathrm{d}u$$

$$= 18\pi - \pi\left[\frac{u^3}{3}\right]\Bigg|_1^3 = 18\pi - \pi(9 - 1/3)$$

3 Problems for Group Work

Problem 1. Find the volume of the solid with rotational symmetry about a vertical axis, whose vertical cross section is bounded by the curves $y = 2x^4$ and $y = 2x^4$ and $y = x^2 + 1$. [Draw picture]

Remark. Multiple approaches are possible here. One is to decompose the solid into two pieces.

Problem 2. Find the volume of a structure extending from $x = 1$ to $x = 4$ whose cross sections perpendicular to the x-axis are the regions below parabolic arcs $y = x - z^2$. [Draw picture]